

UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program • First Semester —
Fall 2021 Sub>ci: Ca lculus Paper: A

Q.1. Solve the
following
questions.

: Rou NO.,
Time: 3 Hrs.

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(615—30)

- (i) Find the limit $\lim_{h \rightarrow 0} \frac{h^2}{1 - \cos h}$.
- (ii) Given $x^3 + y^3 = 3xy^2$. Find $\frac{dy}{dx}$ by implicit differentiation.
- (iii) Let $y = (\ln x)^{\tan x}$. Find $\frac{dy}{dx}$.
- (iv) Evaluate the integral $\int \frac{e^{\tan^{-1} x}}{1 + x^2} dx$.
- (v) Evaluate the integral $\int \sin^4 x \cos^3 x dx$.
- (vi) Find an equation of the plane that passes through the points $C(1, 0, -1)$.

$A(-2, 1, 1), B(0, 2, 3)$

S Qive the following questions.

(3x10=30)

Q.2. Let $f(x) = x^4 - 12x^3$. Find the relative extrema using both first and second derivative tests.

Q.3. Evaluate the improper integral $\int_{-2}^2 \frac{dx}{x^2}$ if it converges.

Q.4. Show that the lines $L_1 : x = 2 - t, y = 2t, z = 1 + t$ and $L_2 : x = 1 + 2t, y = 3 - 4t, z = 5 - 2t$ are parallel, and find the distance between them.

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Fall 2021.

Q 1.

(i) Find the limit $\lim_{h \rightarrow 0} \frac{h^2}{1 - \cosh h}$.

Using L'Hopital rule.

$$\lim_{h \rightarrow 0} \frac{2h}{-(-\sinh h)}$$

Again using L'Hopital rule.

$$= \lim_{h \rightarrow 0} \frac{2}{\cosh h}$$

Put Evaluate limit.

$$= \frac{2}{\cosh 0} = \frac{2}{1} = 2.$$

(ii)
Given $x^3 + y^3 = 3xy^2$. Find $\frac{dx}{dy}$ by implicit differentiation.

$$x^3 + y^3 = 3xy^2$$

$$\frac{d}{dy}(x^3) + \frac{d}{dy}(y^3) = \frac{d}{dy}(3xy^2)$$

$$3x^2 \frac{dx}{dy} + 3y^2 = 3 \frac{dx}{dy} xy^2 + 3x \cdot 2y$$

$$3x^2 \frac{dx}{dy} + 3y^2 = 3y^2 \frac{dx}{dy} + 6xy$$

$$3x^2 \frac{dx}{dy} - 3y^2 \frac{dx}{dy} = 6xy - 3y^2$$

$$\frac{dx}{dy} = \frac{3(2xy - y^2)}{3(x^2 - y^2)} \Rightarrow \frac{dx}{dy} = \frac{2xy - y^2}{x^2 - y^2}$$

DATE: ___/___/___

let $y = (\ln x)^{\tan x}$ (iii). Find $\frac{dy}{dx}$.

$$\left(\frac{dy}{dx} = \frac{d}{dx} ((\ln x)^{\tan x}) \right)^x$$

$$\ln y = \ln (\ln x)^{\tan x}$$

$$\ln y = \tan x (\ln (\ln x)).$$

$$\frac{d}{dx} (\ln y) = \frac{d}{dx} (\tan x) (\ln (\ln x))$$

$$\frac{1}{y} \frac{dy}{dx} = \sec^2 x \ln (\ln x) + \tan x \cdot \frac{1}{\ln x} \cdot \frac{1}{x}$$

$$= \sec^2 x \ln (\ln x) + \tan x \cdot \frac{1}{x \ln x}$$

$$\frac{dy}{dx} = (\ln x)^{\tan x} \left\{ \sec^2 x \ln (\ln x) + \frac{\tan x}{x \ln x} \right\}$$

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$$\int \frac{e^{\arctan x}}{1+x^2} dx.$$

let $z = \arctan x.$

$$dz = \frac{1}{1+x^2} dx.$$

$$= \int e^z dz = e^z$$

$$= e^{\arctan x}.$$

$$= e^{\arctan x}.$$

(v)

Evaluate the integral $\int \sin^4 x \cos^3 x dx.$

$$= \int \cos^2 x \cdot \sin^4 x dx.$$

$$= \int (1 - \sin^2 x) \cos x \sin^4 x dx.$$

let $u = \sin x.$

$$du = \cos x dx.$$

$$= \int (1 - u^2) u^4 du.$$

$$= \int u^4 (1 - u^2) du.$$

$$= \int (u^4 - u^6) du$$

$$= \int u^4 du - \int u^6 du.$$

$$= \frac{u^5}{5} - \frac{u^7}{7} + C$$

$$= \frac{\sin^5 x}{5} - \frac{\sin^7 x}{7} + C.$$

(6)

Find an equation of plane that passes through the points $A(-2, 1, 1)$, $B(0, 2, 3)$ and $C(1, 0, -1)$.

$$\vec{AB} = \langle 0 - (-2), 2 - 1, 3 - 1 \rangle$$

$$= \langle 2, 1, 2 \rangle$$

$$\vec{AC} = \langle 1 - (-2), 0 - 1, (-1) - 1 \rangle$$

$$= \langle 3, -1, -2 \rangle$$

$$\vec{N} = \vec{AB} \times \vec{AC}$$

$$= \begin{vmatrix} i & j & k \\ 2 & 1 & 2 \\ 3 & -1 & -2 \end{vmatrix}$$

$$= i(1(-2) - 2(-1)) - j(2(3) - 2(-2)) + k(2(-1) - 1(3))$$

$$= i(2) - j(10) + k(-5)$$

$$= \langle 2, -10, -5 \rangle$$

Now we have a normal vector $\vec{N} = \langle 2, -10, -5 \rangle$ to the plane and a point $A(-2, 1, 1)$ on the plane.

Using the point-normal form of the equation of a plane:

$$\vec{N} \cdot (\vec{r} - \vec{A}) = 0$$

$$\langle 2, -10, -5 \rangle \cdot (x+2, y-1, z-1) = 0$$

$$2(x+2) - 10(y-1) - 5(z-1) = 0$$

$$2x+4 - 10y+10 - 5z+5 = 0$$

$$2x - 10y - 5z + 19 = 0 \quad \text{Ans.}$$

Solve the following questions.

Q2.

Let $f(x) = x^4 - 12x^3$. Find the relative extrema using both first and second derivative tests.

1st derivative.

$$f(x) = x^4 - 12x^3$$

$$f'(x) = 4x^3 - 12 \cdot 3x^2 \\ = 4x^3 - 36x^2$$

Substitute $f'(x) = 0$.

$$0 = 4x(x - 9)$$

$$x = 0; x = 9$$

2nd derivative.

$$f'(x) = 4x^3 - 36x^2$$

$$f''(x) = 12x^2 - 72x$$

$$f''(0) = 0; f''(9) = 12(9)^2 - 72(9)$$

$$= 972 - 648$$

$$= 324 > 0$$

$f(x)$ is minima on $x = 9$.

When $f''(x) = 0$ then there is no result. So in order to find extremas we have to find using 1st derivative.

As we know that the critical point is zero. So we are supposing two points before and

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after zero.

$$f'(0.1) = 4(0.1)^3 - 36(0.1)^2 = -0.356.$$

$$f'(0.1) = 4(0.1)^3 - 36(0.1)^2 = -0.364.$$

As $f(x)$ is decreasing o.b.s of x . So at $x=0$ $f(x)$ will be maxima.

(Q 3)

Evaluate the improper integral

$$\int_{-2}^2 \frac{1}{x^2} dx \text{ if it converges.}$$

$$= \int_{-2}^0 \frac{1}{x^2} dx + \int_0^2 \frac{1}{x^2} dx.$$

$$= \lim_{t \rightarrow 0} \int_{-2}^t \frac{1}{x^2} dx + \lim_{b \rightarrow 0} \int_b^2 \frac{1}{x^2} dx.$$

$$= \lim_{t \rightarrow 0} \left| \frac{x^{-1}}{-1} \right|_{-2}^t + \lim_{b \rightarrow 0} \left| \frac{x^{-1}}{-1} \right|_b^2.$$

$$= \lim_{t \rightarrow 0} \left| -\frac{1}{x} \right|_{-2}^t + \lim_{b \rightarrow 0} \left| -\frac{1}{x} \right|_b^2.$$

$$= -\frac{1}{0} + \left(\frac{1}{-2}\right) + \left(-\frac{1}{2}\right) + \frac{1}{0} \Rightarrow -\infty + \infty = \infty.$$

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~~Q4.8~~
Show that the lines L_1 and

$$L_1: x=2-t, y=2t, z=1+t$$

and

$$L_2: x=1+2t, y=3-4t, z=5-2t.$$

are parallel, and find the distance between them.

a directional vector of $L_1 = d_1$

$$d_1 = [-1, 2, 1].$$

directional vector of $L_2 = d_2$.

$$d_2 = [2, -4, -2].$$

$$= -2[-1, 2, 1].$$

$$d_2 = -2d_1$$

$$\cos \theta = \frac{u \cdot v}{|u||v|} = \frac{d_1 \cdot d_2}{|d_1||d_2|}$$

$$d_1 \cdot d_2 = [-1, 2, 1] \cdot [2, -4, -2].$$

$$= (-1)(2) + (2)(-4) + (1)(-2).$$

$$d_1 \cdot d_2 = -2 - 8 - 2 = -12.$$

$$|d_1| = \sqrt{(-1)^2 + (2)^2 + (1)^2} = \sqrt{6}.$$

$$|d_2| = \sqrt{2^2 + (-4)^2 + (-2)^2} = \sqrt{24}.$$

$$\cos \theta = \frac{d_1 \cdot d_2}{|d_1||d_2|} = \frac{-12}{\sqrt{6}\sqrt{24}} = \frac{-12}{\sqrt{144}}$$

$$\cos \theta = -1$$

$$\theta = \cos^{-1}(-1)$$

$$\theta = \pi$$